NAD^+

100–300 mg SC 2–3 times weekly for 4–8 weeks; then 2–4 weeks off

NAD^+ is proposed experimentally to support cellular energy metabolism, metabolic resilience, and healthspan through NAD-dependent pathways. [1] NAD^+ is a small dinucleotide coenzyme, not a peptide; it functions as a redox carrier and a substrate for sirtuins, PARPs, and CD38. [1] Oral nicotinamide mononucleotide (NMN) has improved selected metabolic and physical-performance outcomes in aged mice, while human findings with nicotinamide riboside (NR) vary by population and endpoint. [2,3,8] Subcutaneous (SC) NAD^+ has undergone a small preliminary safety study; its proposed functional benefits remain speculative. [9] Combining oral precursors with apigenin is a plausible mechanistic strategy based on preclinical CD38 inhibition. [1,6]

💉 Possibly safe (weak mechanistic / animal evidence / some human use) [9]

🅾️ Inconclusive (Evidence signals are mixed with some showing it doesn't work; precursor studies address different interventions) [2,8,9]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

A pilot preprint included six participants receiving native NAD^+ SC in a study of three daily injections followed by a seven-day washout. Across its small treatment groups, the authors reported injection discomfort but no concerning safety signal; this supports only a preliminary short-term tolerability assessment. [9] In a separate retrospective study, six participants received NAD^+ V on four consecutive days; ALT, AST, and BUN/creatinine did not change significantly at 30 days. These small-sample laboratory observations do not quantify uncommon injury. [4] All six NAD^+ IV recipients reported moderate-to-severe infusion symptoms, including increased heart rate, chest pressure, and gastrointestinal symptoms, resolving when infusion ended. [4]

Functional energy benefits from SC NAD^+ are biologically plausible but remain speculative as a clinical application. [1,9] Oral NR trials provide mixed, population-specific results, rather than a direct test of SC NAD^+. [2,8] Cellular and mitochondrial NAD^+ pools must be considered separately.

Dose and Patterns

A validated SC dose–response relationship for functional energy, endurance, or metabolic flexibility has not been established. The IV pilot studies assess different exposure conditions and cannot define SC efficacy thresholds. [4,5,9]

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The speculative, experimental SC dose and pattern discussed in wellness clinics and peptide communities.

- **100–300 mg SC 2–3 times weekly for 4–8 weeks, followed by 2–4 weeks off.**

Exploratory lower-dose proposals include 25–50 mg and 50–150 mg SC, with their functional effects considered speculative.

Mechanistically, intermittent or context-aligned dosing, such as around exercise or a loading–maintenance sequence, is hypothesized to influence NAD-dependent metabolic signaling while limiting sustained perturbation of homeostasis. [1] Intermittent, pulsed, or cyclic NAD^+ regimens are speculative approaches to preserve responsiveness and limit possible metabolic adaptation.

Pulse dosing: A speculative loading pattern is three SC administrations over 7–10 days, followed by 2–4 weeks off and reassessment. The proposed rationale is transient exposure followed by recovery of metabolic homeostasis, with avoidance of compensatory enzyme regulation as a hypothesis.

The speculative **Intermittent maintenance** pattern is 100–300 mg SC 2–3 times weekly for 4–8 weeks, then 2–4 weeks off. The intended mechanism is periodic pathway engagement rather than continuous exposure.

Oral precursors and supplements

Direct NAD^+ administration introduces the coenzyme by injection, but blood measurements, cellular uptake, and downstream effects depend on route and metabolism. [1,5] IV NAD^+ has produced substantial infusion-related discomfort in small studies. [4,5]

Oral precursors (NR and NMN) avoid injections and feed NAD^+ biosynthetic pathways. [1] Oral NR increased blood NAD^+ in a randomized long-COVID trial, but this biomarker response did not translate into significant randomized improvements in cognition, fatigue, sleep, or mood. [2] Oral products may be more practical for sustained administration.

Apigenin may be paired with NAD^+ precursors as a mechanistic “protect and replenish” hypothesis: apigenin inhibited CD38 and increased NAD^+ in cell and mouse experiments, while NR and NMN supply NAD^+ synthesis pathways. [1,6] Superior clinical efficacy of the combination is speculative.

Extracellular NAD^+ can be metabolized by enzymes including CD38, and precursor utilization involves transport and enzymatic conversion. [1] Plasma-membrane entry, extracellular processing, and intracellular NAD^+ synthesis are distinct determinants of availability. [1]

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Benefits

- Possible support for mitochondrial bioenergetics and NAD-dependent sirtuin activity; functional benefit depends on the experimental context. [1,7]
- Oral NMN, particularly with exercise, improved selected glucose-handling and physical-performance measures in aged mice. [3] In people with peripheral artery disease, oral NR produced a preliminary walking-performance signal. [8]
- Enhanced cellular resilience is a plausible mechanistic objective of supporting NAD-dependent redox and repair pathways. [1]

Applications

- Age-related NAD^+ decline: a mechanistic target for healthspan research. [1]
- Metabolic syndrome: a plausible application supported by preclinical NAD-metabolism research. [1,3,6]
- Mitochondrial dysfunction: a plausible target based on NAD^+ redox biology and mitochondrial transport. [1,7]
- Post-viral syndromes, including long-COVID: oral NR increased blood NAD^+ in a randomized study without significant between-group symptom or cognitive benefits. Differential dropout and exploratory within-group analyses limit interpretation. SC NAD^+ remains a speculative application. [2]

Cautions

- Pregnancy and breastfeeding: precautionary settings; reproductive and developmental safety of repeated SC NAD^+ remains speculative.
- Active malignancy or recent cancer: a precautionary setting because NAD-dependent metabolism and repair can have context-dependent biological consequences; an effect of SC NAD^+ on cancer progression remains speculative. [1]

Side effects

Nausea, vomiting, diarrhea, abdominal cramping, increased heart rate, and chest pressure were reported during infusions in a small retrospective study and resolved after infusion completion. [4] An acute pilot preprint also found poorer infusion tolerability with NAD^+ than NR. [5]

Drivers of Diminished Response and Mitigation Strategies

Classical peptide-receptor desensitization is not the primary model for NAD^+, which is a small metabolic cofactor. [1] Low potential for neutralizing antibodies to the endogenous molecule is a mechanistic inference, separate from inflammatory reactions to a preparation. [10] Compensatory changes in NAD^+ biosynthesis, consumption, redox balance, and NAD-

dependent enzyme activity could plausibly alter responses during sustained exposure. [1] Reduced incremental benefit through such adaptation remains a hypothesis for SC dosing.

Biological Mechanism

NAD^+ is a central redox cofactor and a substrate for NAD-consuming enzymes, including sirtuins, PARPs, and CD38. [1] Increasing NAD^+ availability can support NAD-dependent enzyme activity when substrate availability is limiting; downstream mitochondrial, metabolic, and DNA-repair effects are context-dependent. [1,6] PARPs consume NAD^+ during ADP-ribosylation involved in DNA-damage responses, while CD38 contributes to NAD^+ turnover. [1] Route, precursor identity, metabolism, and dosing pattern can therefore influence compartment-specific exposure.

Blood NAD^+ elevation is not equivalent to sustained NAD^+ elevation in muscle or brain, or to functional improvement. Oral NR raised blood NAD^+ in the long-COVID trial without significant randomized symptom benefit. [2] In the NICE peripheral-artery-disease trial, the small muscle-biopsy subset did not show a significant increase in muscle NAD^+. [8]

Conclusions

NAD-related interventions have produced context-specific benefits rather than a uniform clinical effect. Oral NMN with exercise improved selected metabolic and physical-performance endpoints in aged mice. [3] In long-COVID, oral NR increased blood NAD^+ without significant randomized symptom or cognitive improvements. [2] In the 90-participant NICE trial, oral NR improved six-minute walking distance by 17.6 m relative to placebo at six months but missed statistical significance for definitive efficacy. [8]

Oral NAD^+ precursors plus apigenin remain a plausible mechanistic combination based on NAD^+ synthesis and preclinical CD38 inhibition. [1,6] SC NAD^+ and its proposed cyclic regimens remain experimental approaches whose biological rationale should be distinguished from demonstrated clinical outcomes.

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